

Udyaan

G9 Banana Packaging Development Project

Project Documentation

Project purpose

This Udyaan project is focused on improving the shelf life and packaging suitability of G9 bananas after harvest. It will investigate a suitable bio-plastic material, develop a packaging prototype, test its performance, and assess whether it can become a viable wholesale product for agri-businesses.

1. Problem Statement

G9 bananas begin to lose value after harvest because they continue to ripen and their quality changes during transport and storage. When the fruit reaches the buyer in poor condition, part of the shipment may need to be discounted, rejected, or sold before it reaches its intended market.

The problem affects several groups. Agri-businesses and produce buyers face the direct commercial impact because they purchase, handle, transport, and sell the bananas. Farmers are affected when produce is rejected or receives a lower value. Consumers are affected when bananas have a shorter usable life or when there is limited information available about whether produce is organic or chemical-free.

Current packaging is generally treated as a container rather than as a system designed around the specific needs of the fruit. It may not adequately manage the conditions that accelerate ripening, and may not provide a clear trust feature for organic or chemical-free produce. Conventional packaging can also create sustainability concerns depending on the material used and its end-of-life treatment.

The problem persists because cold storage and refrigerated transport are not always practical or affordable for every movement of fresh produce. Packaging therefore has to perform effectively under ordinary storage and transport conditions. A solution also needs to be affordable and practical for the businesses that would purchase it.

The project will investigate whether a bio-plastic packaging design can help maintain the quality of G9 bananas for longer and communicate a credible organic or chemical-free trust feature. If successful, the solution could reduce avoidable product loss, improve the value of shipments, support better buyer confidence, and create a wholesale packaging opportunity for agri-businesses.

Primary customer and stakeholders

Group	Relationship to the project
Agri-business or produce buyer, such as Oye Desi	Primary potential customer and wholesale buyer
Farmers	Suppliers who may benefit from fewer rejected or downgraded bananas
Transporters and handlers	Users who load, stack, move, and unload the packages
Retailers and final buyers	Users who need bananas to remain presentable and usable
Consumers	Beneficiaries of better quality and clearer product information
Project team and academic advisors	Developers, reviewers, and validators of the prototype

2. Work Breakdown & Milestones

The project is divided into milestones. Each milestone has a defined output and an exit criterion so that progress can be reviewed objectively.

Milestone 1: Problem and user confirmation

Tasks

- Confirm the G9 banana use case.
- Confirm the main buyer and user groups.
- Document the transport, storage, and shelf-life problems.
- Confirm the required organic or chemical-free trust feature.

Output: Locked problem statement and stakeholder summary.

Exit criterion: A reader outside the team can identify the affected groups, the current problem, the reason it persists, and the intended change.

Milestone 2: Requirements definition

Tasks

- Define the required packaging functions.
- Separate confirmed requirements from features requiring validation.
- Define the basic acceptance tests for shelf life, package usability, and trust communication.

Output: Requirements sheet with status for each requirement.

Exit criterion: The team has an agreed list of requirements and knows how each one will be reviewed.

Milestone 3: G9 banana and packaging research

Tasks

- Review the storage needs of G9 bananas.
- Study factors related to ripening, ventilation, moisture, and temperature.
- Review relevant packaging formats and their limitations.

Output: G9 research brief and packaging reference summary.

Exit criterion: The design decisions are supported by a documented understanding of the banana and its storage and transport conditions.

Milestone 4: Bio-plastic material research

Tasks

- Identify suitable bio-plastic material options.
- Compare food-contact safety, strength, flexibility, moisture resistance, availability, cost, and end-of-life characteristics.

- Consider whether each material is suitable for the planned packaging structure and trust feature.

Output: Materials comparison and preliminary material recommendation.

Exit criterion: The team has selected a material direction for prototyping or documented why further research is required.

Milestone 5: Packaging concept development

Tasks

Output: Packaging concept sheet with sketches, dimensions, material direction, and trust-feature concept.

- Develop the package structure and dimensions.
- Plan ventilation, access, storage, and usability features.
- Define how the package will communicate the organic or chemical-free trust feature.

Exit criterion: The concept is sufficiently defined to build a prototype.

Milestone 6: Prototype development

Tasks

- Build the first physical prototype or small prototype batch.
- Record material quantities, construction method, and estimated production cost.
- Check whether the package can be loaded, stacked, moved, opened, and displayed practically.

Output: Physical prototype and prototype build record.

Exit criterion: A usable prototype exists and can be tested under defined conditions.

Milestone 7: Testing and validation

Tasks

- Compare packaged bananas with the current or ordinary packaging approach where possible.
- Observe shelf-life indicators over the test period.
- Record shelf-life, temperature, moisture, and usability observations.
- Review the trust feature for clarity and credibility.
- Document limitations and required design changes.

Output: Validation memo with observations, photographs or records where available, and recommended changes.

Exit criterion: The team can state which requirements the prototype meets, which require iteration, and which remain unverified.

Milestone 8: Commercial and handover review

Tasks

- Estimate material and production cost.
- Define an initial wholesale pricing approach.
- Gather feedback from the intended business user where available.
- Record the budget and reimbursements.
- Recommend the next build, pilot, or research step.

Output: Commercial review and final handover package.

Exit criterion: The project has a completed prototype, supporting records, a basic business model, and a clearly stated next step.

3. Revenue Opportunities

The initial model is deliberately simple: develop a usable package and sell completed packages wholesale to businesses that handle fresh produce. Additional revenue streams can be considered after the first product and validation milestones are complete.

Still, more in-depth research is needed.

Opportunity	Offering	Paying customer	Monetisation model	Required proof
Wholesale packaging sales	Completed packages supplied in batches	Agri-businesses, distributors, and produce sellers	Product sale	Prototype, cost estimate, and buyer feedback
Custom packaging development	Adaptation of the package for a buyer's supply-chain needs	Agri-business or produce brand	Service fee plus development cost	Validated base design and documented requirements
Packaging supply agreement	Recurring supply of packages after successful adoption	Regular wholesale buyer	Contracted product sale or subscription-style supply agreement	Repeat-use evidence and reliable production process
Design or process licensing	Permission to use a validated packaging design or process	Packaging manufacturer or produce company	Licensing fee or royalty	Distinctive design, documented process, and legal review
Trust-feature integration	Packaging information or traceability feature integrated into supply	Produce brand or agri-business	Service fee, product margin, or commission depending on the system	Working feature and credible verification process

The first commercial focus should remain wholesale packaging sales. Service, licensing, recurring supply, and commission-based opportunities should only be pursued when the underlying product and process have been validated.

4. Team Structure & Resourcing

The following is a proposed team structure for completing the developmental block. Names and final headcounts can be assigned by the team and confirmed with Shradha mam.

College department or function	Role	Proposed headcount	Skills required	Main responsibility
Biochemistry	Produce and materials research lead	1-2	Food science, biological materials, observation and documentation	G9 banana research, shelf-life factors, and material suitability
CNMS or equivalent science department	Testing and validation lead	1-2	Experimental design, measurement, data recording, basic analysis	Test planning, usability observations, and validation memo
Design or fabrication support	Packaging design and prototype lead	1-2	Product design, prototyping, dimensions, construction methods	Package structure, physical prototype, and iteration
Business or management function	Commercial and customer research lead	1	Market research, costing, communication, basic pricing	Buyer identification, wholesale model, costing, and feedback
Project coordination	Project lead and documentation owner	1	Planning, coordination, writing, milestone tracking	Schedule, records, handover package, and authority communication

Resourcing principles

- One person may hold more than one role if the available team is small.
- Each task should have one accountable owner, even when several people contribute.
- Department mapping should reflect the actual college structure before final submission.
- Specialist testing or material advice should be requested where the internal team cannot verify a requirement safely.

5. Timeline

The project uses estimated development durations. The dates are not payment guarantees and may change after material availability, testing access, or advisor feedback is confirmed.

Milestone	Estimated duration	Dependency	Critical path
Problem and user confirmation	1-2 days	None	Yes
Requirements definition	1-2 days	Milestone 1	Yes
G9 banana and packaging research	2-4 days	Milestone 1	Yes
Bio-plastic material research	3-7 days	Milestones 1-2	Yes
Packaging concept development	2-4 days	Milestones 2-4	Yes
Prototype development	3-7 days	Milestone 5 and material availability	Yes
Testing and validation	5-14 days	Milestone 6	Yes
Commercial and handover review	2-4 days	Milestone 7	No

Scenario estimates

- **Best case:** approximately 10 days, if materials, facilities, and testing access are immediately available.
- **Realistic case:** approximately 20 days, allowing time for research, prototype construction, testing, and one revision.
- **Worst case:** up to 2 months, if material sourcing, testing, advisor review, or prototype iteration takes longer than expected.

The critical path runs through requirements, research, material selection, concept development, prototype development, and validation. Commercial review and final documentation can proceed alongside later testing where appropriate.

6. Social Impact

The project is intended to reduce avoidable loss and improve confidence in fresh produce packaging. Its impact should be measured through the prototype and pilot evidence rather than through unsupported estimates.

Intended effects

- Longer period in which bananas remain acceptable for sale or consumption.
- Better visibility of organic or chemical-free product information.
- Reduced dependence on packaging that is unsuitable for the intended use or difficult to manage at end of life.
- Better value retention for agri-businesses and fewer rejected or downgraded bananas for farmers.

Measurement framework

Impact area	Baseline	Proposed measure
Shelf life	Bananas stored using the current or ordinary packaging method	Days until the product reaches the agreed unacceptable-quality threshold
Buyer confidence	Existing packaging information	User feedback on clarity and credibility of the organic or chemical-free trust feature
Material suitability	Current packaging material and disposal practice	Material safety, performance, cost, availability, and end-of-life comparison

The baseline should be recorded before final testing. If a controlled comparison is not possible, the document should identify the result as an observation rather than a confirmed impact measurement.

7. Final Deliverable

Primary deliverable

A usable physical prototype, or small prototype batch, of the bio-plastic packaging for G9 bananas.

Supporting handover package

The prototype will be handed over with:

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| 1. Locked problem statement and stakeholder summary. | 2. Requirements sheet covering food safety, temperature effects, bio-plastic suitability, moisture resistance, shelf-life impact, cost, manufacturability, end-of-life or combustion effects, and other relevant packaging factors, with each requirement marked as confirmed, requiring further research, or requiring validation. |
| 3. G9 banana and packaging research brief. | 4. Bio-plastic materials comparison and recommendation. |
| 5. Packaging concept sheet with dimensions and trust-feature design. | 6. Prototype build record and estimated unit cost. |
| 7. Testing and validation memo. | 8. Budget and reimbursement record. |
| 9. Basic wholesale business model. | 10. Recommended next build, pilot, or research step. |

Recipient

The final package will be presented to the project authorities, advisors, and intended business stakeholders, including relevant Oye Desi representatives where applicable.

Acceptance criteria

The final deliverable will be considered complete when:

- A physical prototype or small prototype batch has been produced.
- The package is designed specifically for G9 bananas.
- The selected or proposed bio-plastic material is documented with supporting research.
- The organic or chemical-free trust feature is included as a defined part of the package.
- Shelf-life, material, and package performance have been tested or clearly marked as requiring further research or validation.
- The prototype's limitations are documented honestly.
- A preliminary cost and wholesale sales model are included.
- The work is supported by a clear validation record and recommended next step.

A patent or other intellectual-property filing is not a required final deliverable. It may be considered later if the prototype includes a distinctive, potentially protectable feature and an appropriate review confirms that further action is worthwhile.